



PLC 101

The Brain of Industrial Automation

Ruggedized Industrial Computers Built for 24/7 Operation

Scan Cycle • Hardware • Ladder Logic • Compact & Modular • IEC 61131-3

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Welcome to PLC 101

Programmable Logic Controllers (PLCs) run almost every modern factory, water plant, elevator, conveyor, packaging line and CNC machine on Earth. They are the rugged, deterministic computers that bridge the physical world of sensors and motors with the logical world of code — and they have been doing it reliably since 1968, when Dick Morley invented the first one to replace the relay logic in a General Motors plant.

This guide takes you from the basic question — what actually is a PLC? — through the scan cycle, the hardware, the ladder logic, and the practical choice between compact bricks and modular racks. Every concept is grounded in real maintenance and commissioning practice, the way it shows up on a shop floor.

What you will learn

Module	Topic	Key takeaway
1	What Is a PLC?	How software replaced wall-sized relay panels.
2	The Continuous Scan Cycle	Read → Execute → Update in 10–15 ms forever.
3	Essential Hardware Building Blocks	CPU, PSU, I/O modules and the backplane bus.
4	Mastering Ladder Logic	XIC, XIO, OTE — and the seal-in trick.
5	Compact vs Modular Configurations	When a brick is enough, when you need a rack.

i INFO A note on standards

Throughout this guide, code examples and instruction names follow the IEC 61131-3 standard, with parallel Allen-Bradley / Siemens shorthand where it differs. The underlying concepts apply to every major brand — Siemens, Allen-Bradley, Mitsubishi, Omron, Schneider, ABB, LS, Delta — even if the editor screens look different.

1 What Is a PLC?

A Programmable Logic Controller is an industrial computer hardened for the worst the shop floor can offer: temperature extremes, vibration, electrical noise, dust, humidity, voltage spikes, and 24/7/365 uptime expectations. It runs a single purpose-built program in a deterministic loop, watching its inputs and driving its outputs in milliseconds.

Before the PLC arrived, the same job was done by literal walls of electromechanical relays wired together. Changing the sequence of a machine meant rerouting hundreds of physical wires — a job that took days and stopped production. Today the same change takes minutes on a laptop.

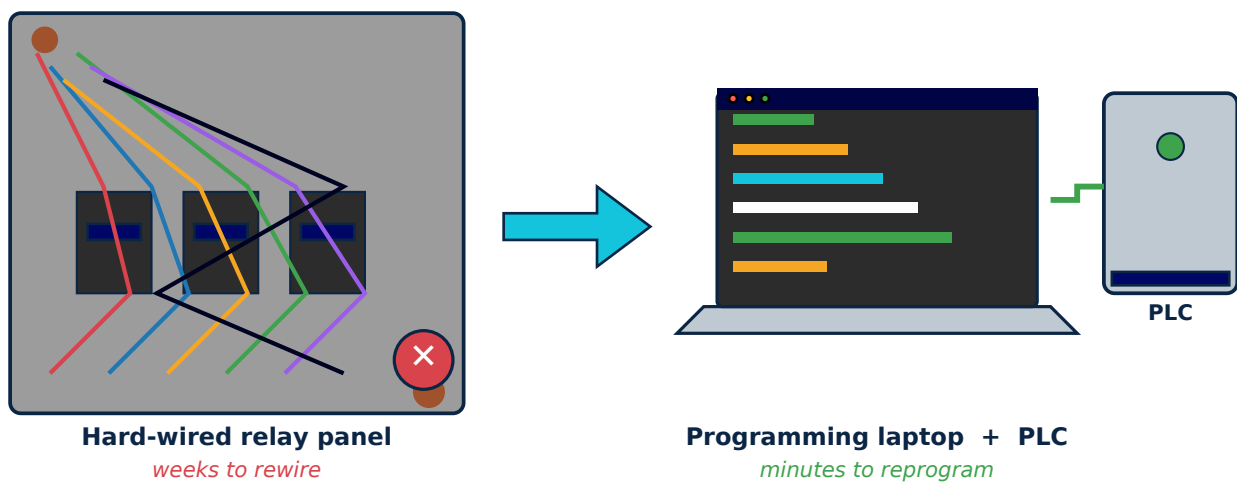


Figure 1.1 — Hard-wired relay logic (replaced) versus a modern PLC programmed from a laptop

Why PLCs replaced relay logic

Aspect	Relay panel (pre-1968)	Modern PLC
Logic changes	Rewire physical relays — days of downtime	Edit code, download, restart — minutes
Footprint	Entire wall of a cabinet	Single DIN-rail brick or rack
Reliability	Contacts wear out, pit, weld	Solid-state, MTBF > 100 000 hours
Diagnostics	Read the schematic and probe with a meter	Online monitoring, force tags, datalogs
Networking	None	Ethernet/IP, PROFINET, Modbus, OPC-UA
Cost per I/O	High — relays, terminals, wires, labor	Falling every year; commodity I/O cards

2 The Continuous Scan Cycle

The single most important thing to understand about a PLC is that it does not execute its program top-to-bottom once like a PC script. It runs a deterministic loop, called the scan cycle, over and over again at high speed — typically every 10-15 milliseconds for a mid-size program.

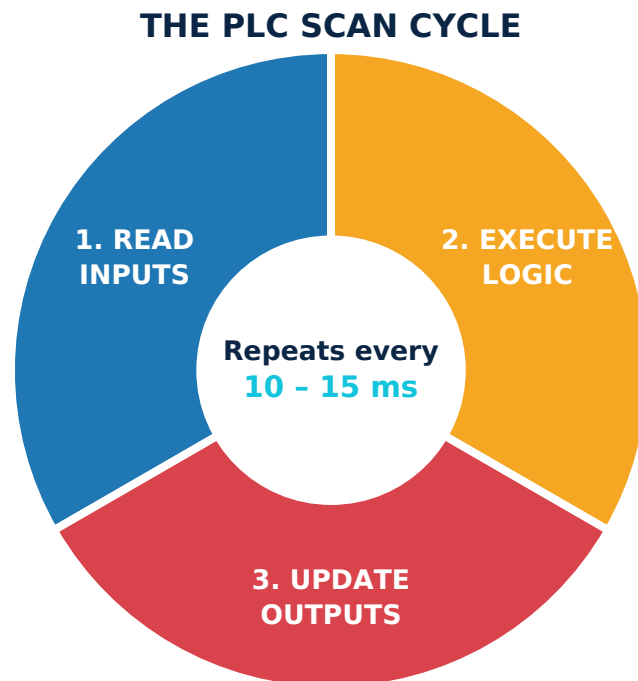


Figure 2.1 — The three phases of the PLC scan cycle

Phase	What happens
1. Input Scan	All physical input states (sensors, buttons, limit switches) are sampled simultaneously and copied to the input image table in memory.
2. Program Execution	The CPU executes the user program top-to-bottom, left-to-right, using the image table values — not the live wires.
3. Output Update	All output image bits are written to the physical output cards in one atomic pass; the actual relay or transistor switches happen now.
(Housekeeping)	Communications, diagnostics, force-table service, and watchdog refresh all happen between cycles.

✓ TIP Why the image table matters

Because the program executes from a frozen snapshot of the inputs, every rung sees the same input states for the entire scan. This is what makes ladder logic deterministic — without it, a sensor flickering mid-program could give two rungs contradictory readings. It also means an input shorter than one scan time can be missed; high-speed events need interrupts or high-speed counter modules.

3 Essential Hardware Building Blocks

Every PLC system — from a 100-USD micro PLC to a six-rack process controller — is built from the same three functional blocks: a CPU that runs the program, a power supply that feeds the electronics, and I/O modules that connect the PLC to the real world.

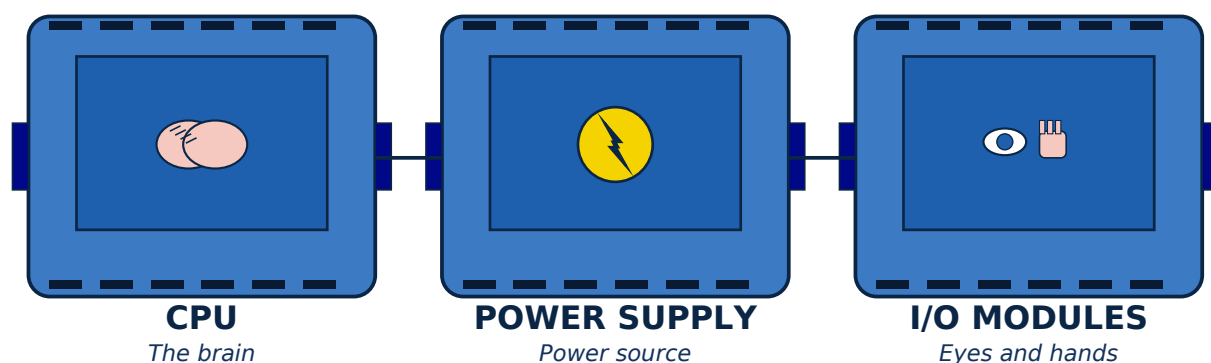


Figure 3.1 — The three essential PLC hardware blocks: CPU, Power Supply, and I/O modules

Block	Role	Key specs to check
CPU	Runs the user program, manages memory, services communications.	Scan time, instructions/sec, RAM size, comms ports.
Power Supply	Converts plant power (120/230 VAC or 24 VDC) into clean rail voltages.	Output current (A), backplane wattage budget, redundancy.
Digital Input (DI)	Reads ON/OFF signals: buttons, limit switches, prox sensors, relays.	Voltage (24 VDC / 120 VAC), isolation, response time.
Digital Output (DO)	Drives ON/OFF loads: contactors, solenoids, indicators, small valves.	Type (relay / transistor / triac), current per point, fusing.
Analog Input (AI)	Reads continuous values: temperature, pressure, flow, level.	Resolution (12/16-bit), range (4–20 mA, 0–10 V), isolation.
Analog Output (AO)	Drives continuous setpoints: VFD speed, valve %, recorder.	Resolution, output type (4–20 mA / 0–10 V), drive current.
Comm modules	Ethernet/IP, PROFINET, Modbus TCP/RTU, DeviceNet, AS-i, IO-Link.	Protocol, isolated/non-isolated, ports, max nodes.

! WARNING Sinking vs sourcing — get it right the first time

A sinking (NPN) input expects the sensor to pull the input pin to 0 V. A sourcing (PNP) input expects the sensor to push 24 V to the pin. Wire a sourcing sensor to a sinking card and nothing will work; worse, you may exceed the absolute-max input voltage. Always read the I/O card's wiring diagram before pulling the first wire.

4 Mastering Ladder Logic

Ladder Logic (LD) is the most widely used of the five IEC 61131-3 programming languages because it visually resembles the relay schematics it replaced. Each rung is read left-to-right: input conditions on the left, output coils on the right. Power rails on either side mirror the L and N of a real control circuit.

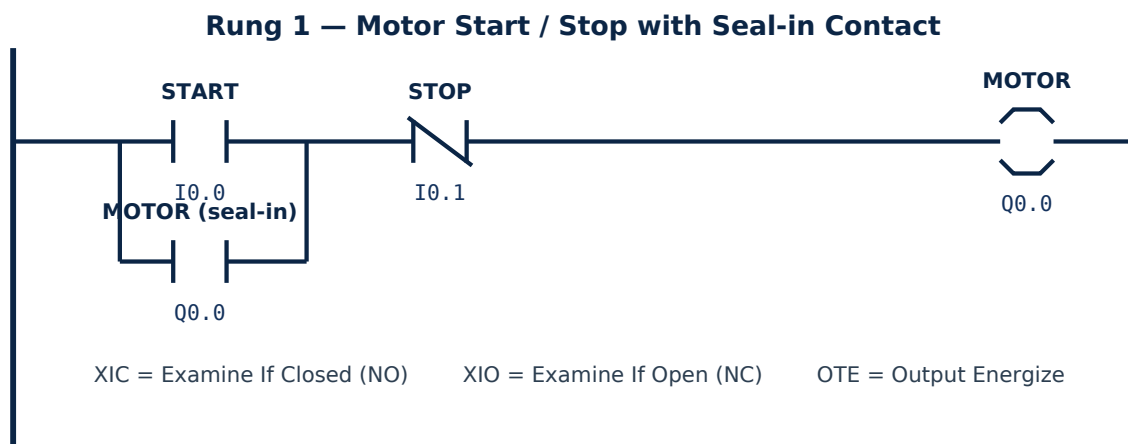


Figure 4.1 — Classic motor start / stop ladder rung with a sealing contact

Pressing START (I0.0) energizes the MOTOR coil (Q0.0); the coil's seal-in contact in parallel with START keeps the motor running after the operator releases the button, until STOP (I0.1) opens the NC contact and drops Q0.0.

The five IEC 61131-3 languages

Code	Name	Best for
LD	Ladder Diagram	Discrete logic, interlocks, safety chains — most popular.
FBD	Function Block Diagram	Continuous control (PID), signal flow, process automation.
SFC	Sequential Function Chart	Step-by-step processes: batch reactors, machine sequences.
ST	Structured Text	Math, loops, string handling — Pascal-like syntax.
IL	Instruction List	Assembly-style; rarely used today, deprecated in newer IEC.

✓ TIP Read every rung from left to right

A rung evaluates left to right, and the leftmost FALSE contact stops evaluation of that branch. Place your most likely-FALSE contacts first to save scan time on huge programs — and always put your safety (E-stop, guard door, motor overload) contacts as master control relay rungs at the very top of the program.

5

Compact vs Modular Configurations

PLCs come in two physical families. A compact PLC packs CPU, power, and a fixed set of I/O into one DIN-rail brick — cheap, simple, and ideal for small machines. A modular PLC uses separate plug-in cards on a rack or backplane, letting you mix and match exactly the I/O the application needs and expand later.

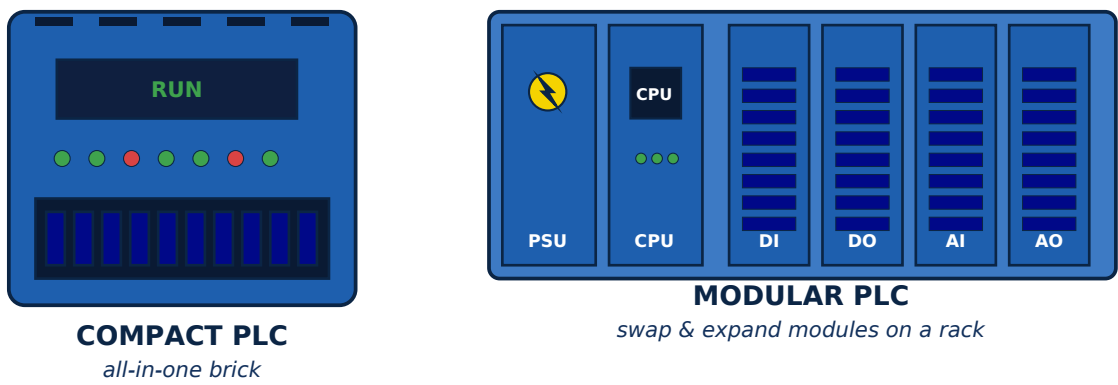


Figure 5.1 — Compact 'brick' PLC versus a modular rack-style PLC

Aspect	Compact PLC	Modular PLC
I/O count	Typically 8–60 fixed points	32 to 8 000+, scalable by adding modules
Mix of I/O types	Fixed at purchase	Any mix — DI, DO, AI, AO, comms, motion
Expansion	Limited (1–3 expansion bricks max)	Slide-in cards on the backplane bus
Footprint	One DIN-rail brick	19-inch rack or backplane in a cabinet
Typical brands	Siemens S7-1200, AB Micro850, Mitsubishi FX	Siemens S7-1500, AB ControlLogix, Mitsubishi iQ-R
Best for	Standalone machines, OEM equipment, retrofits	Process lines, multi-cell plants, complex motion / safety

i INFO 20% rule for sizing

Whatever I/O count you calculate from the P&ID, add at least 20% spare before specifying the PLC. Modifications during commissioning, late operator requests, and field-discovered sensors will eat that margin every single time. Spare cards are always cheaper than re-cabineting a live plant.



Worked Example: Sizing a PLC for a Conveyor

A typical small conveyor with start/stop control, photoelectric package detection, and VFD speed control — let's walk through the I/O count and PLC selection.

I/O type	Device	Count
Digital Input (24 VDC)	START button, STOP button, E-stop, photoeye, motor overload, jam sw.	6
Digital Output (relay)	Run lamp, Fault lamp, Audible alarm, VFD enable	4
Analog Output (4-20 mA)	VFD speed reference	1
Analog Input (4-20 mA)	Conveyor weigh-cell feedback	1
Communications	Ethernet/IP to HMI & SCADA	1
Subtotal	Before spare margin	12 pts + 1 net
+ 20% spare	Always size up; future-proof	≈ 14 pts
Recommended PLC	Compact PLC with 14+ I/O and Ethernet (S7-1200, Micro850, FX5U etc.)	OK

Pseudo-ladder for the conveyor (Rung overview)

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Rung 1 (Safety): E-Stop_OK AND Overload_OK → Master_Control_Relay
Rung 2 (Start): MCR AND ( START OR Motor_Run ) AND NOT STOP → Motor_Run
Rung 3 (VFD): Motor_Run → VFD_Enable; set AO speed_ref from HMI
Rung 4 (Counter): Photoeye rising edge → Pkg_Count++
Rung 5 (Alarm): Jam_Sensor OR Weigh_OverRange → Fault_Lamp, Alarm_Horn
    
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Self-Check Quiz

Five short questions to lock the concepts in. Try to answer first, then read the explanation in italics.

1. Why does a PLC use an input image table instead of reading wires directly during program execution?

Determinism. Every rung in a single scan sees the same input snapshot, so two rungs cannot get contradictory readings from a flickering sensor. The trade-off is that any pulse shorter than one scan time can be missed — that's what high-speed counter modules and interrupts are for.

2. A 'seal-in' contact is placed where, and what does it do?

It's an NO contact of the output coil, wired in parallel with the START button. Once the coil is energized, the seal-in keeps it energized after the START button is released — until a STOP (NC) opens the rung and drops the coil.

3. Which IEC 61131-3 language is best for a PID temperature loop?

Function Block Diagram (FBD). PID is shipped as a function block by every major vendor; FBD lets you wire its inputs (PV, SP, tuning) and outputs (CV) visually like a signal-flow diagram.

4. A sensor labeled 'PNP, 24 VDC' is wired to an input card labeled 'NPN / sinking'. What happens?

Nothing — or worse. A PNP sensor sources +24 V to its output; a sinking card expects the sensor to pull its terminal to 0 V. The signal never reaches the threshold, so the input never sees TRUE. Always match sensor type to card type (or use a card that supports both).

5. You have a small standalone packaging machine with 16 DI, 8 DO, no analog. Compact or modular PLC?

Compact. The I/O count is fixed and well within a single-brick PLC's capability. Modular is overkill — more expensive, takes more panel space, more SKUs to stock. Save modular for expandable process lines.

i INFO Where to go next

If you scored 4 / 5 or higher, you are ready for our deeper PLC courses on Siemens TIA Portal, Rockwell Studio 5000, and IEC structured text. Visit ayetechub.com for the full curriculum and downloadable practice projects.



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About the author

Awet G. Nway is a Robotics Maintenance and PLC Operations specialist with hands-on experience in industrial automation, machine foundation supervision, production-and-maintenance team leadership, and HVAC/MEP systems. He is the founder of AYE Tech Hub, an engineering education and services platform serving learners and businesses across Ethiopia and beyond.

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